SUPERMARKET SALES ANALYTICS

Project Report: Dataset, Dashboard Features, Analysis and Findings

Prepared for: Power BI / Data Analytics Project

Dataset period: 1 January 2019 – 30 March 2019

This report explains the dataset fields, dashboard features, KPI calculations, key findings and limitations.

1. Executive Summary

This project analyses 1,000 supermarket sales transactions recorded across three branches from 1 January to 30 March 2019. It uses a Power BI dashboard to summarise revenue, transaction volume, product-line performance, branch performance, payment methods and customer patterns.

The dashboard transforms raw transaction records into interactive visualisations and key performance indicators (KPIs), helping users understand what happened in the available data. The findings are descriptive and apply to this dataset and time period.

2. Project Objectives

• Understand the dataset structure and the meaning of each field.

• Calculate important measures such as total revenue, transaction count, average order value, quantity sold, gross income and average rating.

• Compare sales by month, branch, product line and payment method.

• Explore customer type and gender categories in the recorded transactions.

• Use interactive visuals, slicers, tooltips and cross-filtering to make the report easier to explore.

• Explain findings and limitations without making claims the dataset cannot support.

3. Dataset Overview

Dataset: Supermarket Sales, commonly distributed as supermarket_sales.csv.

Rows: 1,000 transaction records. Columns: 17 fields.

Date range: 1 January 2019 to 30 March 2019.

Branches: A, B and C. Payment methods: Cash, Credit card and Ewallet.

Product lines: Food and beverages; Sports and travel; Electronic accessories; Fashion accessories; Home and lifestyle; Health and beauty.

Each row represents a transaction. Some columns identify or describe the transaction, while others contain numerical measures.

Currency note: the source dataset is commonly labelled in Myanmar kyat (MMK). Confirm the currency and display label in your Power BI report before presenting it.

4. Explanation of All 17 Dataset Fields

Invoice ID — Unique transaction identifier. Use it to count transactions; do not add invoice IDs together.

Branch — Branch code A, B or C. Used to compare sales between branches.

City — City associated with the branch. Useful for location-based grouping in this dataset.

Customer type — Customer category, recorded as Member or Normal. Useful for comparing transaction patterns by customer type.

Gender — Customer gender category recorded in the dataset (Female or Male). It supports descriptive counts, not broad conclusions about all customers.

Product line — Product category purchased, such as Food and beverages or Health and beauty. Used to compare category revenue and volume.

Unit price — Price of one unit of the purchased product before multiplying by quantity.

Quantity — Number of product units purchased in the transaction.

Tax 5% — Sales tax amount charged at the dataset's 5% rate.

Total — Total bill amount for the transaction, including the tax in this dataset. Use it for total customer-bill revenue if that is your chosen definition.

Date — Date of the transaction. Used to group sales by day, month or period.

Time — Time of the transaction. Can be used to study busy times of day if time visuals are included.

Payment — Method used to pay: Cash, Credit card or Ewallet.

cogs — Cost of goods sold: the recorded cost associated with the products sold in the transaction.

gross margin percentage — Gross margin percentage provided in the source. It is effectively constant in this dataset, so it is not useful for comparing records.

gross income — Gross income amount recorded for the transaction. It is not net profit because all operating expenses are not provided.

Rating — Customer rating recorded for the transaction. Average rating can be used as a simple customer-feedback indicator.

5. Data Preparation and Quality Checks

Check column data types in Power Query: Date should be a date; Time should be a time; Quantity should be a whole number; prices, tax, total, cogs and gross income should be numeric; Invoice ID should be text.

Check for missing values, duplicate records, spelling differences and invalid numbers. The previously reviewed summary reported no missing cells and no duplicate rows, but repeat these checks against the exact CSV you use.

Make sure month names are sorted by month number. Otherwise, Power BI may sort month labels alphabetically instead of chronologically.

Format monetary measures consistently to two decimal places, quantity as a whole number and rating to a sensible number of decimal places.

If the report grows beyond this three-month sample, consider creating a calendar table for reliable time-based analysis.

6. Dashboard Features Explained

KPI cards — Large summary cards display Total Revenue, Total Transactions, Average Order Value, Total Quantity Sold, Total Gross Income and Average Rating. They allow viewers to understand the overall results quickly.

Monthly revenue trend — A line or column chart compares revenue across January, February and March 2019. It highlights movement during the available period, not a full-year trend.

Branch comparison — A bar or column chart compares revenue for Branches A, B and C and shows which branch contributes the most.

Product-line analysis — A ranked bar chart compares revenue across the six product categories and identifies the highest- and lowest-revenue categories.

Payment analysis — A pie/donut or bar chart compares Cash, Ewallet and Credit card. The title should state whether it shows transaction count, revenue or percentage share.

Customer analysis — Charts compare Member and Normal customers and/or the Female and Male records in the dataset. These are sample descriptions, not evidence of general population behaviour.

Rating analysis — A card shows the average rating. An optional bar chart can compare rating by branch or product line.

Slicers — Interactive filters may be added for date, branch, city, product line, customer type, gender and payment method. Selecting a slicer updates connected visuals.

Tooltips — Hovering over a visual can show additional details, such as revenue, quantity, transaction count or average rating.

Cross-filtering/highlighting — Selecting a bar or chart segment can filter or highlight related visuals, helping users investigate patterns.

Labels and formatting — Clear titles, readable labels, consistent currency formatting and restrained colours improve readability.

Page layout/navigation — Place KPI cards at the top, then trend and comparison charts, followed by customer/payment insights. A clean layout helps when presenting the project.

7. KPI Definitions and Reported Values

Total Revenue = SUM(Total): 322,966.75.

Total Transactions = DISTINCTCOUNT(Invoice ID): 1,000.

Average Order Value = Total Revenue ÷ Total Transactions: approximately 322.97.

Total Quantity Sold = SUM(Quantity): 5,510 units.

Total Gross Income = SUM(gross income): approximately 15,379.37.

Average Rating = AVERAGE(Rating): approximately 6.97.

Total Tax = SUM(Tax 5%): approximately 15,379.37.

These figures come from the previously reviewed dataset summary. Recalculate and verify them in Power BI after importing the exact CSV. Total includes tax; gross income should not be described as net profit.

8. Key Findings

Monthly revenue: January — 116,291.87; February — 97,219.37; March — 109,455.51. January is highest and February lowest among these three months; revenue rises again in March.

Branch revenue: A — 106,200.37; B — 106,197.67; C — 110,568.71. Branch C is highest; Branches A and B are almost equal.

Product-line revenue: Food and beverages — 56,144.84; Sports and travel — 55,122.83; Electronic accessories — 54,337.53; Fashion accessories — 54,305.90; Home and lifestyle — 53,861.91; Health and beauty — 49,193.74. Food and beverages is highest and Health and beauty is lowest by revenue.

Transaction count by payment: Ewallet — 345; Cash — 344; Credit card — 311. Ewallet leads transaction count by only one transaction over Cash.

Revenue by payment: Cash — 112,206.57; Ewallet — 109,993.11; Credit card — 100,767.07. Cash contributes the highest revenue total.

Customer type: Member — 501 transactions; Normal — 499. The counts are almost evenly split.

Gender: Female — 501 records; Male — 499. These counts are also almost evenly split.

9. Suggested Power BI DAX Measures

```
Total Revenue = SUM('supermarket_sales'[Total])

Total Transactions = DISTINCTCOUNT('supermarket_sales'[Invoice ID])

Average Order Value = DIVIDE([Total Revenue], [Total Transactions], 0)

Total Quantity Sold = SUM('supermarket_sales'[Quantity])

Total Gross Income = SUM('supermarket_sales'[gross income])

Average Rating = AVERAGE('supermarket_sales'[Rating])

Total Tax = SUM('supermarket_sales'[Tax 5%])
```

Replace 'supermarket_sales' with the actual table name in your Power BI model. Column names must match the imported names exactly.

10. Limitations and Cautions

The dataset covers only three months in 2019. It cannot support reliable year-on-year analysis or conclusions about annual seasonality.

Only three branches and a limited set of locations are represented, so results should not be generalised to every supermarket.

Total includes the 5% tax. State clearly whether the dashboard’s revenue measure means the customer’s total bill or pre-tax sales.

Gross income is not net profit. Additional operating costs and expenses would be needed to calculate net profit.

Gross margin percentage is effectively constant in this dataset and does not help compare transactions.

A higher revenue value may reflect price, quantity, or both. The charts show associations and totals, not necessarily causes.

Category and customer labels reflect the source dataset’s collection and classification choices.

11. Conclusion

The Supermarket Sales Analytics dashboard makes the 1,000 transaction records easier to understand by summarising revenue, sales volume, branch performance, product categories, payment choices and customer patterns. In the reviewed results, Branch C has the highest revenue, Food and beverages is the leading product line, Cash produces the highest payment-method revenue, and Ewallet has the most transactions by a narrow margin.

This project demonstrates data preparation, KPI calculation, visualisation and interactive analysis in Power BI. Interpret the findings within the limits of the three-month dataset.

12. Suggested Project Files

• Power BI report (.pbix), source CSV (where sharing is permitted), dashboard screenshot, this report in PDF/Word format, and a README explaining the project and key findings.